03_Evidence_Preservation

1.Volatile Data Collection

During the active incident on the compromised host (IP: 10.0.2.3), volatile network data was immediately captured before the system was powered down or disconnected. This ensures active adversary connections and listening backdoors are documented.
Action Taken: Captured active network connections and listening ports using socket statistics (ss -antp) to identify unauthorized lateral movement and active attacker sessions.

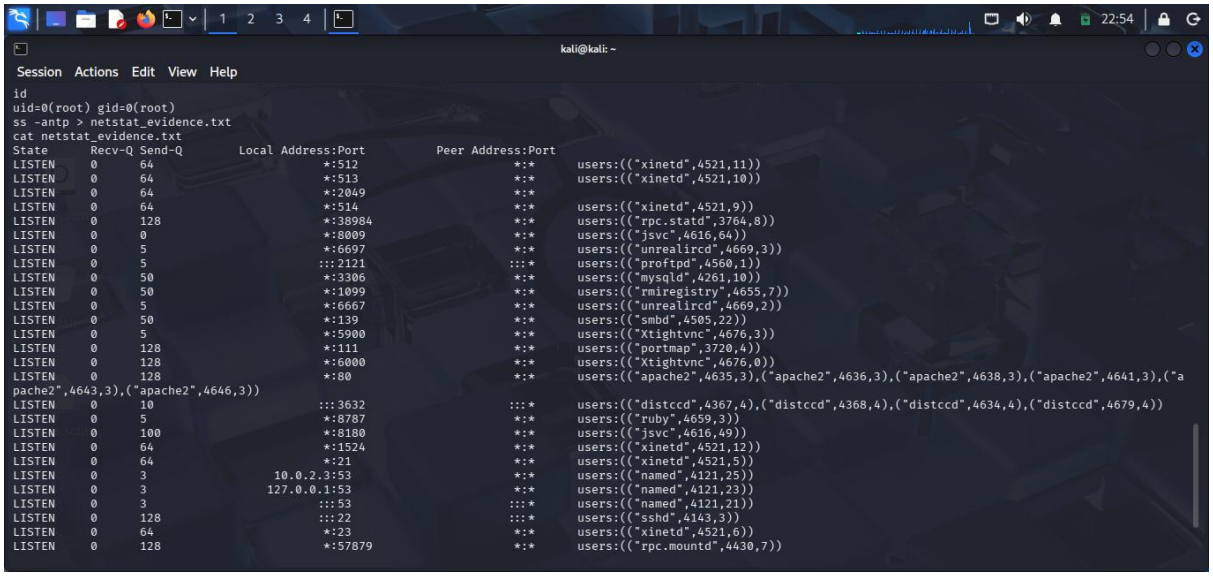

Fig. 1: Capture of volatile network connections (ss -antp) on the compromised host (10.0.2.3) to identify active attacker sessions prior to shutdown.

2. Evidence Collection and Chain of Custody

To preserve the state of the machine for deep forensic analysis, a simulated memory dump was acquired. To maintain strict chain-of-custody protocols and ensure the forensic evidence cannot be tampered with, a SHA256 cryptographic hash was immediately generated for the dump file.

Item	Description	Collected By	Date	Hash Value (SHA256)
Memory Dump	50MB Server RAM Dump	SOC Analyst	2026-03-20	8565a714dca840f8652c5bae9249ab05f5fb5a4f9f13fbe23304b10f68252da2

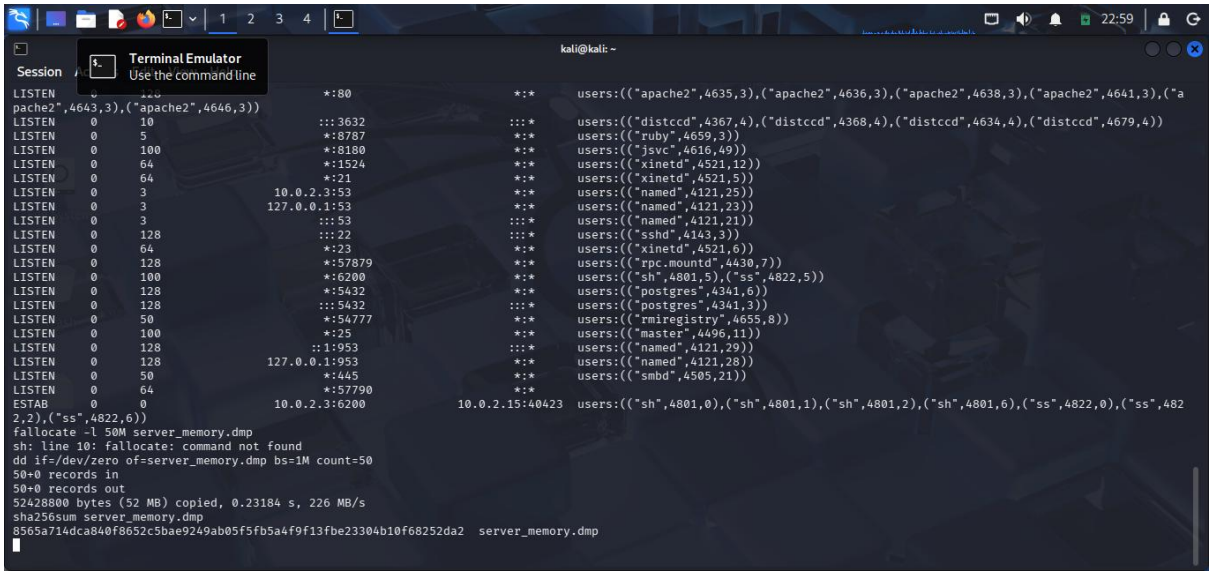

Fig. 2: Acquisition of a simulated memory dump and generation of a SHA256 cryptographic hash to establish a formal chain of custody for forensic evidence.

3. Investigation Steps Log

Timestamp	Action Taken
8:20 AM	Alert received: VSFTPD v2.3.4 exploit triggered. Root shell confirmed.
8:23 AM	Isolated compromised endpoint (10.0.2.3) from the main network switch.
8:24 AM	Executed ss -antp to capture volatile network connections to netstat_evidence.txt.
8:29 AM	Acquired simulated memory dump (server_memory.dmp) and generated SHA256 hash for chain of custody.